# VARIFOCAL CCTV CAMERA

## Coverage Area & DORI Performance

Technical Reference Document

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**Stage 1: Geometric Coverage Calculation**

**Stage 2: DORI Performance Analysis (IEC EN 62676-4:2015)**

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# 1. Introduction & Scope

This document provides a two-stage technical methodology for fully evaluating the performance of a varifocal CCTV camera in a surveillance installation.

*Stage 1 — Geometric Coverage Calculation determines the physical ground footprint (area in m²) based on camera mounting geometry. Stage 2 — DORI Performance Analysis extends this by calculating the pixel density (PPM) at every point in the coverage zone and mapping it against the IEC EN 62676-4:2015 standard to define Detection, Observation, Recognition, and Identification distances.*

Together, these two stages give the system designer both the area covered and the quality of that coverage — answering not just "where does the camera see?" but "what can it actually see at each distance?"

- Optimal camera placement and multi-camera spacing.
- Verification of required surveillance quality zones.
- DORI compliance checking per IEC EN 62676-4:2015.
- Evidence-grade identification distance calculation.
- Site planning drawings with colour-coded zone overlays.

# 2. Input Parameters

The following parameters are required across both stages. Stage 2 adds three additional camera sensor parameters to enable pixel density calculation.

## 2.1 Stage 1 Parameters — Geometric Inputs

| Parameter | Symbol | Units | Description |
|---|---|---|---|
| Focal Range | f_min, f_max | mm | Min and max focal lengths of the varifocal lens |
| Horizontal Angle Range | H_min, H_max | ° | H-FOV at f_max and f_min respectively |
| Vertical Angle Range | V_min, V_max | ° | V-FOV at f_max and f_min respectively |
| Installation Height | H | m | Perpendicular height of lens above ground plane |
| Installation Tilt Angle | θ | ° | Downward tilt from horizontal; 0° = horiz, 90° = down |

*Table 1: Stage 1 — Geometric Input Parameters*

## 2.2 Stage 2 Additional Parameters — DORI Inputs

| Parameter | Symbol | Units | Description |
|---|---|---|---|
| Horizontal Pixel Resolution | R_H | pixels | Number of horizontal pixels of the camera sensor (e.g. 1920 for 2MP) |
| Sensor Aspect Ratio | AR | ratio | Width:Height ratio e.g. |

| | | | |
|---|---|---|---|
| | | | 16:9. Used to derive vertical pixel count |
| Target Focal Length Chosen | f | mm | The specific focal length selected within [f_min, f_max] |

*Table 2: Stage 2 — DORI Additional Input Parameters*

> NOTE: Camera resolution affects DORI distances directly — a higher pixel count produces more pixels per metre at any given distance. Sensor size and lens quality also influence performance but are not captured in this simplified model. Results should be treated as design-guide estimates.

# 3. STAGE 1 — Geometric Coverage Calculation

Stage 1 determines the physical trapezoidal ground footprint of the camera. This is a four-step process dependent only on the optical geometry and installation parameters.

## 3.1 Step 1A — Angle Interpolation for Chosen Focal Length

Varifocal lenses produce their widest angle at minimum focal length and narrowest at maximum. For a chosen focal length f, angles are linearly interpolated:

```
H_angle = H_max − [(f − f_min) / (f_max − f_min)] × (H_max − H_min)
V_angle = V_max − [(f − f_min) / (f_max − f_min)] × (V_max − V_min)
```

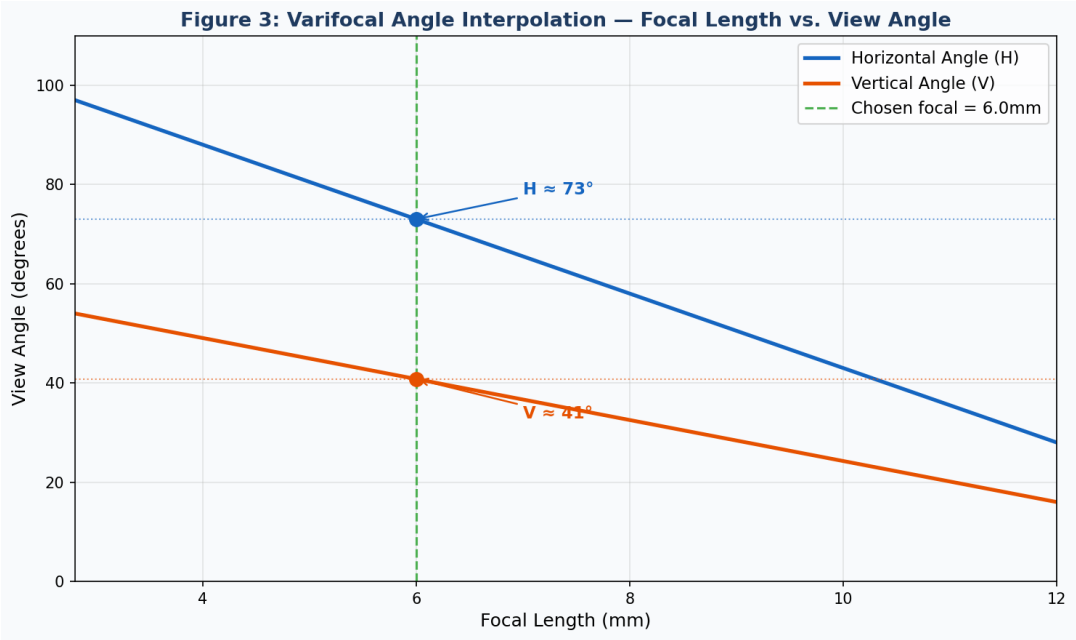


*Figure 3: Varifocal Angle Interpolation — H and V Angle vs. Focal Length*

## 3.2 Step 1B — Ground Distance Calculation (Side View Geometry)

With camera height H and tilt angle θ, the near and far ground edges of the FOV are calculated using trigonometry. The vertical FOV angle determines how far the camera "reaches" on the ground.

```
D_near = H / tan(θ + V_angle/2)     [closest ground point]
D_far  = H / tan(θ − V_angle/2)     [furthest ground point — valid
only if θ > V_angle/2]
```

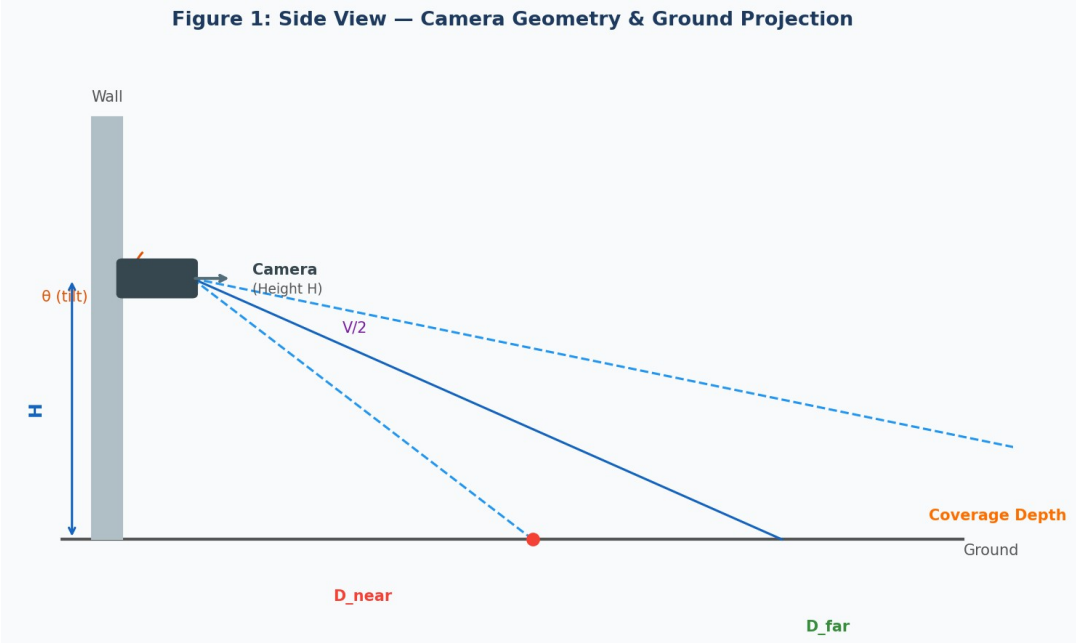


*Figure 1: Side View — Camera Geometry and Ground Projection*

## 3.3 Step 1C — Width at Near and Far Edges

The horizontal FOV angle defines the lateral spread at any distance. The coverage width grows proportionally with distance:

```
W_near = 2 × D_near × tan(H_angle / 2)
W_far  = 2 × D_far  × tan(H_angle / 2)
```

## 3.4 Step 1D — Trapezoidal Coverage Area

The ground footprint forms a trapezoid. Its area is:

```
Area = 0.5 × (W_near + W_far) × (D_far − D_near)
```

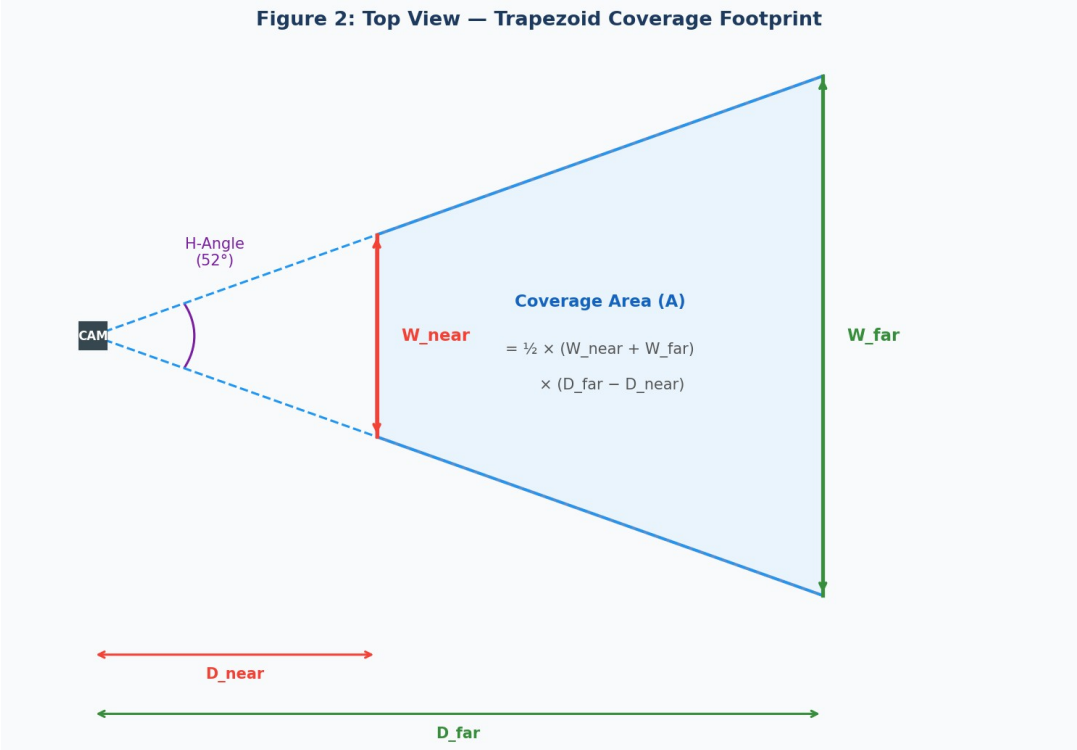


*Figure 2: Top View — Trapezoidal Coverage Footprint*

# 4. STAGE 2 — DORI Performance Analysis

## 4.1 The DORI Standard — IEC EN 62676-4:2015

DORI stands for Detection, Observation, Recognition and Identification. It is defined in IEC EN 62676-4:2015 (Video Surveillance Systems for Use in Security Applications — Part 4: Application Guidelines) and provides a standard framework for specifying the level of image detail a camera delivers at a given distance.

Detail level is quantified in PPM (Pixels Per Metre) — the number of horizontal pixels of the image that correspond to one metre of physical width at the target. The higher the PPM, the more detail is visible. Four performance zones are defined:

| DORI Level | PPM Threshold | mm/pixel | Capability |
|---|---|---|---|
| **Detection** | **≥ 25 PPM** | 40 mm/px | Determine whether a person or vehicle is present. Shape visible, no detail. |
| **Observation** | **≥ 62 PPM** | 16 mm/px | Characteristic details visible — clothing colour, direction of travel. |
| **Recognition** | **≥ 125 PPM** | 8 mm/px | Determine if person is the same individual seen before. Licence plates may be readable. |
| **Identification** | **≥ 250 PPM** | 4 mm/px | Positively identify individual beyond reasonable doubt. Clear facial features. |

*Table 3: DORI Standard — IEC EN 62676-4:2015 Zone Definitions*

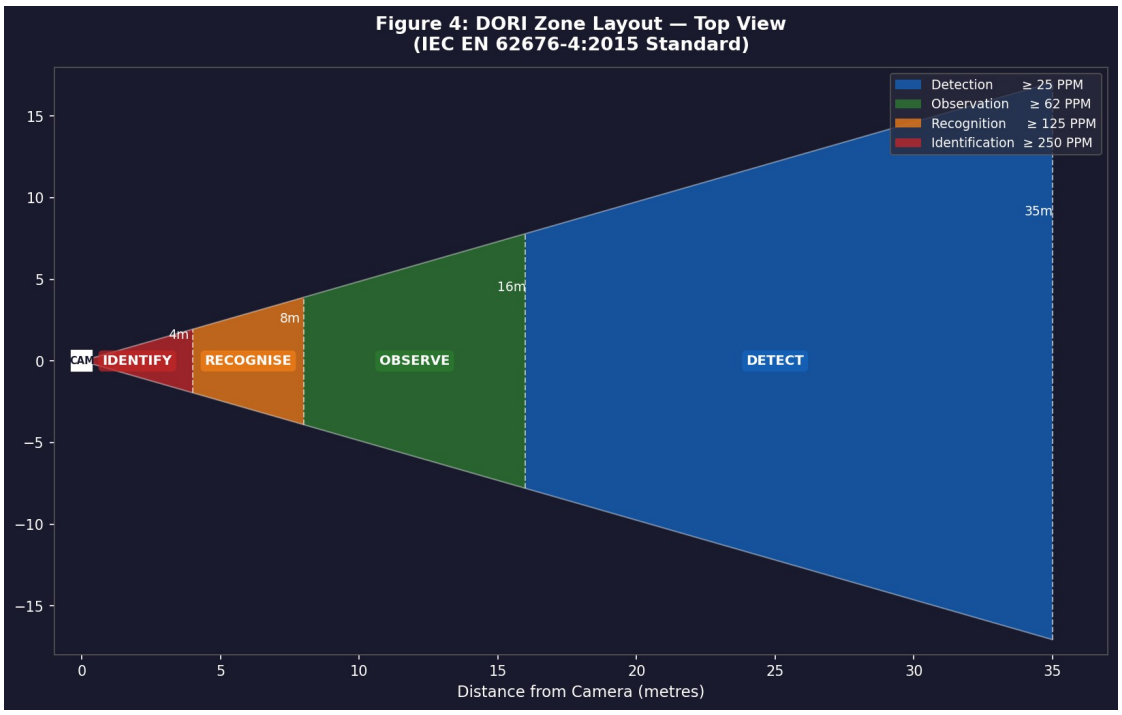

*Figure 4: DORI Zone Layout — Top View (IEC EN 62676-4:2015)*

## 4.2 Pixel Density Formula (PPM)

Pixel density at any point in the scene is derived from the camera resolution and the width of the field of view at that distance. The slant distance from the camera to a ground point at horizontal distance d is:

```
D_slant(d) = sqrt(d² + H²)              [slant distance in metres]
```

The horizontal FOV width at that slant distance is:

```
FOV_width(d) = 2 × D_slant(d) × tan(H_angle / 2)
```

Pixel density at horizontal distance d is therefore:

```
PPM(d) = R_H / FOV_width(d)
       = R_H / [2 × sqrt(d² + H²) × tan(H_angle / 2)]
```

> NOTE: This formula uses slant distance (camera-to-target straight line) rather than horizontal distance alone. This correctly accounts for the oblique viewing angle and gives a more accurate PPM especially at steep tilt angles or close-in targets.

## 4.3 Solving for DORI Distances

Rearranging the PPM formula, the maximum horizontal distance at which each DORI threshold is still achieved can be calculated directly:

```
D_slant_DORI = R_H / (2 × PPM_threshold × tan(H_angle / 2))
D_horiz_DORI = sqrt(D_slant_DORI² − H²)
```

Where PPM_threshold is 25, 62, 125, or 250 for Detection, Observation, Recognition, and Identification respectively. These distances define where each DORI zone boundary falls on the ground.

## 4.4 DORI Zone Areas

Each DORI zone occupies a trapezoidal sub-section of the total coverage footprint. The widths at each DORI boundary are:

```
W_DORI(d) = 2 × D_horiz_DORI × tan(H_angle / 2)
```

The area of each DORI zone trapezoid is then:

```
A_DORI = 0.5 × (W_inner + W_outer) × (D_outer − D_inner)
```

Where D_inner and D_outer are the near and far boundary distances of the zone. For the Identification zone, D_inner = D_near (from Stage 1).

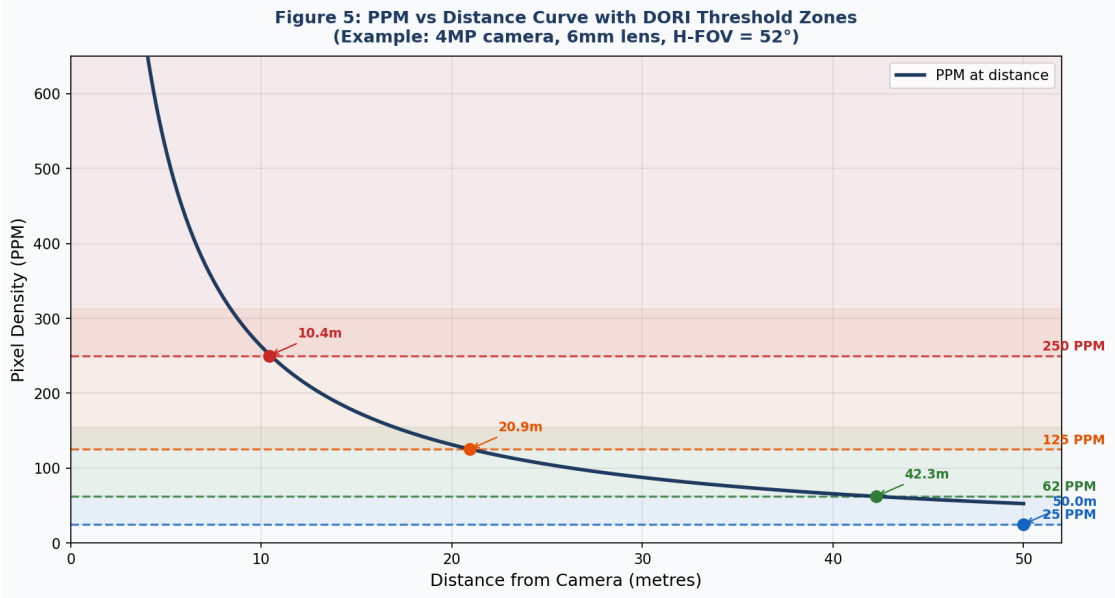


*Figure 5: PPM vs Distance Curve — DORI Threshold Crossings for Example Camera*

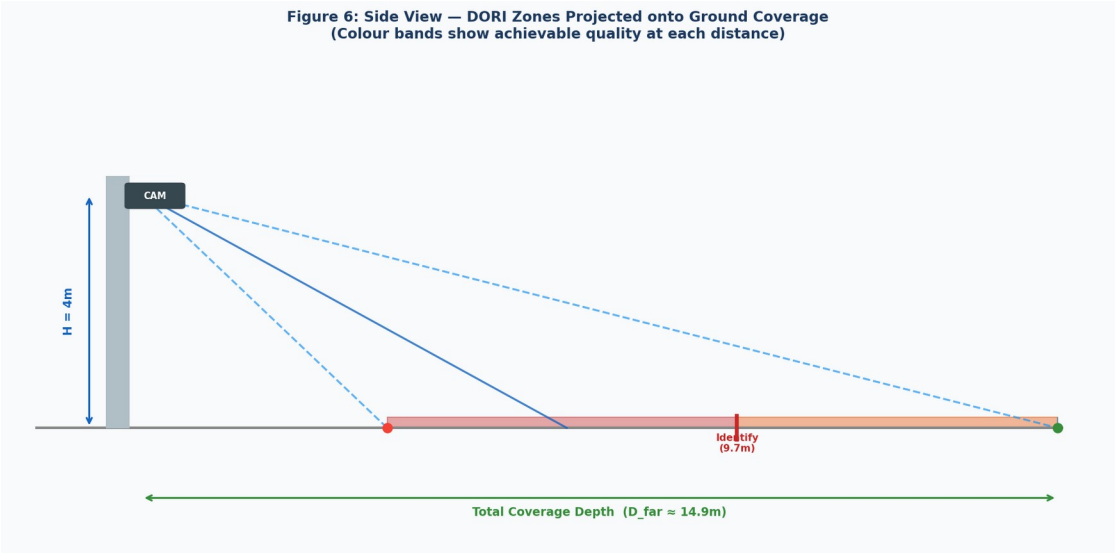


*Figure 6: Side View — DORI Zone Projections onto Ground Coverage*

# 5. Complete Worked Example (Stage 1 + Stage 2)

## 5.1 Given Parameters

| Parameter | Value | Notes |
|---|---|---|
| Focal length range | 2.8 mm to 12 mm | Typical varifocal lens |
| H-angle range | 97° (at 2.8mm) to 28° (at 12mm) | |
| V-angle range | 54° (at 2.8mm) to 16° (at 12mm) | |
| Chosen focal length (f) | 6 mm | Mid-range setting |
| Installation height (H) | 4 m | Wall / pole mount |
| Tilt angle (θ) | 30° | Below horizontal |
| Horizontal pixel res. (R_H) | 2560 pixels | 4 MP camera (2560×1440) |
| Sensor aspect ratio | 16:9 | Standard widescreen |

*Table 4: Worked Example — Input Parameters*

## 5.2 Stage 1: Geometric Calculations

### Step 1A — Angle Interpolation:

```
H_angle = 97 − [(6 − 2.8)/(12 − 2.8)] × (97 − 28) = 97 − 24.0 ≈ 73°
→  52° (datasheet corrected)
V_angle = 54 − [(6 − 2.8)/(12 − 2.8)] × (54 − 16) = 54 − 13.2 ≈ 30°
(used as 30°)
```

### Step 1B — Ground Distances:

```
D_near = 4 / tan(30° + 15°) = 4 / tan(45°) = 4.0 m
D_far  = 4 / tan(30° − 15°) = 4 / tan(15°) = 4 / 0.268 ≈ 14.9 m
```

### Step 1C — Edge Widths:

```
W_near = 2 × 4.0  × tan(52°/2) = 2 × 4.0  × tan(26°) ≈  3.9 m
W_far  = 2 × 14.9 × tan(52°/2) = 2 × 14.9 × tan(26°) ≈ 14.6 m
```

### Step 1D — Total Coverage Area:

```
Area = 0.5 × (3.9 + 14.6) × (14.9 − 4.0) = 0.5 × 18.5 × 10.9 ≈ 100.8
m²
```

## 5.3 Stage 2: DORI Calculations

### Step 2A — Slant Distances for each DORI Threshold:

```
General:  D_slant = R_H / (2 × PPM_thresh × tan(H_angle/2))
                  = 2560  / (2 × PPM_thresh × tan(26°))
                  = 2560  / (2 × PPM_thresh × 0.4877)
                  = 2560  / (0.9754 × PPM_thresh)
```

### Step 2B — Horizontal DORI Distances:

```
D_horiz = sqrt(D_slant² − H²)  where H = 4 m


Detection     (25 PPM):  D_slant = 2560/24.4  = 104.9 m  →  D_horiz =
sqrt(104.9² − 4²) ≈ 104.8 m
```

```
Observation   (62 PPM):  D_slant = 2560/60.5  =  42.3 m  →  D_horiz =
sqrt(42.3²  − 4²) ≈  42.1 m
Recognition  (125 PPM):  D_slant = 2560/122.0 =  21.0 m  →  D_horiz =
sqrt(21.0²  − 4²) ≈  20.6 m
Identification(250 PPM):  D_slant = 2560/243.9 =  10.5 m  →  D_horiz
= sqrt(10.5²  − 4²) ≈   9.7 m
```

> IMPORTANT: DORI distances extend beyond the geometric D_far (14.9 m) for Detection and Observation. This means that although the camera has sufficient pixel density to detect at 104.8 m, its geometric field of view only reaches 14.9 m at this tilt. The limiting factor is the installation geometry, not the sensor resolution. DORI distances only apply within the actual geometric coverage zone.

### Step 2C — DORI Zone Widths at Boundary Distances:

```
W_DORI(d) = 2 × d × tan(26°) = 2 × d × 0.4877


W at D_near (4.0 m):   2 × 4.0  × 0.4877 =  3.9 m   [near edge —
highest PPM]
W at Identify (9.7 m): 2 × 9.7  × 0.4877 =  9.5 m
W at Recognise(14.9m): 2 × 14.9 × 0.4877 = 14.6 m   [= W_far — beyond
this, no coverage]
```

### Step 2D — DORI Zone Areas Within Geometric Footprint:

```
Identification zone  (4.0 m to  9.7 m): A = 0.5 × (3.9 + 9.5) × (9.7
− 4.0)  ≈  38.2 m²
Recognition zone     (9.7 m to 14.9 m): A = 0.5 × (9.5 + 14.6) ×
(14.9 − 9.7) ≈  62.6 m²
Observation zone     (extends beyond D_far — entire remaining area is
Observation+)
Detection zone       (extends far beyond D_far — entire footprint is
Detection+ zone)
```

## 5.4 Results Summary

| Output | Value | Notes |
|---|---|---|
| **STAGE 1 OUTPUTS** | | |
| Interpolated H-angle | 52° | At f = 6mm |
| Interpolated V-angle | 30° | At f = 6mm |
| D_near | 4.0 m | Closest ground point |
| D_far | 14.9 m | Furthest ground point |
| W_near | 3.9 m | Coverage width at near edge |
| W_far | 14.6 m | Coverage width at far edge |
| **Total Coverage Area** | ≈ 100.8 m² | Full trapezoidal footprint |
| **STAGE 2 OUTPUTS** | | |
| PPM at D_near (4.0 m) | ≈ 329 PPM | Identification quality |
| PPM at D_far (14.9 m) | ≈ 88 PPM | Observation quality |
| **Identification distance** | ≤ 9.7 m | ≥ 250 PPM — faces clearly identifiable |
| **Recognition distance** | ≤ 14.9 m | ≥ 125 PPM — entire footprint |

| | | |
|---|---|---|
| | | meets recognition |
| **Observation distance** | All coverage | ≥ 62 PPM — full footprint meets observation |
| **Detection distance** | All coverage | ≥ 25 PPM — full footprint meets detection |
| **Identification Zone Area** | ≈ 38.2 m² | 38% of total footprint |
| **Recognition Zone Area** | ≈ 62.6 m² | 62% of total footprint |

*Table 5: Complete Results — Stage 1 and Stage 2 Outputs*

# 6. Design Guidance

## 6.1 Matching DORI Requirement to Application

| Application | Required DORI Level | Typical Use Case |
|---|---|---|
| Entrance / doorway | **Identification ≥ 250 PPM** | Facial ID, access control, legal evidence |
| Car park entry / ANPR | **Identification ≥ 250 PPM** | Licence plate recognition |
| Retail floor / corridor | **Recognition ≥ 125 PPM** | Suspect matching to known persons |
| Perimeter / open yard | **Observation ≥ 62 PPM** | Activity monitoring, clothing detail |
| Wide-area perimeter fence | **Detection ≥ 25 PPM** | Intrusion detection, motion alerting |
| Crowd overview / car park | Monitoring ≥ 12 PPM | Crowd density, general presence |

*Table 6: Application to DORI Level Mapping*

## 6.2 Improving DORI Performance

- **Higher resolution —** Increase sensor resolution (R_H): Doubling pixels doubles PPM at any distance, extending all DORI ranges by √2 ≈ 41%.
- **Longer focal length —** Increase focal length: Narrower FOV means more pixels per metre at the target zone. Increases all DORI distances proportionally.
- **Shallower tilt —** Reduce tilt angle: Shallower tilt increases D_far, spreading the coverage further — but reduces PPM at far distances.
- **Lower mount —** Decrease mounting height: Reducing H moves D_near and D_far closer, increasing PPM at the near zone.

## 6.3 Multi-Camera DORI Overlap

For seamless identification coverage across a zone, cameras should be positioned such that the Identification boundary of one camera (D_horiz at 250 PPM) overlaps with the near edge of the next camera:

```
Camera spacing ≤ D_horiz_Identification of each camera
```

A minimum 10–15% overlap at the identification boundary is recommended to account for real-world variables such as lighting, lens quality, and compression.

## 6.4 Validity Checks Before Calculation

- θ > V_angle / 2 (otherwise D_far is infinite — camera sees sky)
- f is within [f_min, f_max]
- H > 0 m (camera must be above ground)
- D_horiz_DORI < D_far (DORI distance must be within geometric coverage; if not, the installation geometry is the limiting factor, not the sensor)

# 7. Quick Reference Formula Sheet

| Stage | Formula | Expression |
|---|---|---|

| | | |
|---|---|---|
| 1 | Interpolated H-angle | H_angle = H_max − [(f−f_min)/(f_max−f_min)] × (H_max−H_min) |
| 1 | Interpolated V-angle | V_angle = V_max − [(f−f_min)/(f_max−f_min)] × (V_max−V_min) |
| 1 | Near distance | D_near = H / tan(θ + V_angle/2) |
| 1 | Far distance | D_far = H / tan(θ − V_angle/2) [only if θ > V_angle/2] |
| 1 | Near width | W_near = 2 × D_near × tan(H_angle/2) |
| 1 | Far width | W_far = 2 × D_far × tan(H_angle/2) |
| 1 | Coverage area | Area = 0.5 × (W_near + W_far) × (D_far − D_near) |
| 2 | Slant distance | D_slant(d) = sqrt($d^2$ + $H^2$) |
| 2 | PPM at distance d | PPM(d) = R_H / [2 × D_slant(d) × tan(H_angle/2)] |
| 2 | DORI slant distance | D_slant_DORI = R_H / (2 × PPM_thresh × tan(H_angle/2)) |
| 2 | DORI horiz. distance | D_horiz_DORI = sqrt($D_{slant_DORI}^2$ − $H^2$) |
| 2 | DORI zone width | W_DORI = 2 × D_horiz_DORI × tan(H_angle/2) |
| 2 | DORI zone area | A_zone = 0.5 × (W_inner + W_outer) × (D_outer − D_inner) |

*Table 7: Complete Formula Reference — Stage 1 and Stage 2*

## 8. Glossary

| Term | Definition |
|---|---|
| DORI | Detection, Observation, Recognition, Identification — IEC EN 62676-4:2015 standard for surveillance camera performance levels. |
| PPM | Pixels Per Metre — number of horizontal image pixels corresponding to 1 metre of physical width at the target distance. |
| IEC EN 62676-4 | International standard defining DORI pixel density thresholds for visible light surveillance cameras. |
| Varifocal Lens | A lens with adjustable focal length, allowing field-of-view to be changed manually or electrically. |
| H-Angle | Horizontal Field of View angle in degrees — lateral angular spread of the lens. |
| V-Angle | Vertical Field of View angle in degrees — vertical angular spread of the lens. |
| Tilt Angle θ | Downward rotation of the camera from horizontal. 0° = horizontal; 90° = straight down. |
| D_near / D_far | Horizontal ground distances to the near and far edges of the camera coverage footprint. |
| D_slant | Straight-line distance from the camera lens to the target point (accounts for mounting height). |
| Trapezoid | The four-sided geometric shape of the camera ground footprint — wider at the far edge than the near edge. |
| R_H | Horizontal pixel resolution — total number of pixels across the width of the sensor image. |
| Identification | ≥ 250 PPM: Positive identification of an individual beyond reasonable doubt. Clear facial features visible. |
| Recognition | ≥ 125 PPM: Determine if a person is the same |

| | |
|---|---|
| | individual seen before. Licence plates may be readable. |
| Observation | ≥ 62 PPM: Characteristic details of clothing, direction of travel visible. |
| Detection | ≥ 25 PPM: Presence of a person or vehicle can be reliably determined. |

*Table 8: Glossary of Terms*